# Ankush Jain

Backend Engineer

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## SUMMARY


Backend engineer with five years building distributed systems for regulated healthcare and imaging — EHR integration, cloud infrastructure, access control and AI pipelines. Comfortable owning a system end to end: high- and low-level design, implementation, security hardening, and the CI/CD that ships it.


## EXPERIENCE

**Apra Labs** May 2023 – Present

*Software Engineer*

Engineering services company. Work is delivered for external clients across healthcare, medical devices and imaging; clients are not named. Also contributed to Apra Labs' own products and open source.

**Clinical Trial Management Platform** (Nov 2024 – Present)

- Built the integration hub's base adapter modules, turning each new EHR integration from a bespoke build into a repeatable pattern; implemented the first adapter as the reference implementation
- Designed and shipped a full role-based access control system across two applications, covering admin and basic access - admin-only endpoint restriction, per-role response filtering, and pre-authorization checks
- Re-architected inter-service connectivity on managed pub/sub with async subscriptions, dead-letter handling and retry enhancement
- Cut the backend deployment pipeline from 26 min 33 s to 18 min 07 s (-32%), shortening every developer's feedback loop
- Identified unnoticed idle cloud spend and shut the service down, saving the project $200+ and prompting a billing-review habit

**Regenerative Medicine Data Platform** (May 2026 – Present)

- Modelled clinical protocols and surveys on FHIR resources with OMOP/LOINC terminology, and implemented change logs purely through FHIR provenance
- Drove requirements with the client directly - converting whiteboard sessions and slide feedback into wireframed workflows and user stories

**Document AI / Handwriting Recognition Pipeline** (May 2025 – Nov 2025)

- Reduced the deployable image from 19 GB to 5.28 GB (-72%) through deliberate packaging design
- Designed the codebase so the third-party model packages and deploys with the system and its licence can be swapped on expiry, instead of requiring a rebuild
- Produced the high- and low-level design independently with minimal supervision

**Headless Video Recording Platform** (Apr 2024 – Oct 2024)

- Wrote the Node backend from scratch and connected it to Redis, the C++ core and the frontend, giving the system one consistent source of truth
- Took the recorder from an unstable state to a dependable one, debugging intermittent segmentation faults in frame decoding across edge hardware

**AR Surgical Navigation** (Aug 2023 – Apr 2024)

- Identified the signed distance field computation as the pipeline's bottleneck and devised a compute-once-and-reuse technique, saving over 80% of the time it took per frame
- Onboarded two junior engineers and led the project through a stretch when senior engineers were unavailable, having become its most familiar contributor

## SKILLS

**Languages:** Python, C++, Java, Kotlin, Go, JavaScript/Node.js, TypeScript
**Backend & messaging:** Spring Boot, Ktor, REST APIs, Server-Sent Events (SSE), RabbitMQ, Google Cloud Pub/Sub, Redis, Message brokers, Event-driven architecture, Microservices
**Cloud & DevOps:** Google Cloud Platform (GCP), Cloud Run, Terraform, Docker, Docker Compose, CI/CD pipelines, Infrastructure as code (IaC), AWS (exposure)
**Healthcare & standards:** HL7 FHIR, HAPI FHIR, OMOP, LOINC, SMART on FHIR, EHR integration, Clinical trial systems, HIPAA-adjacent data handling
**Security:** Role-based access control (RBAC), Authentication & authorization, OAuth / token-based auth, Microsoft Graph API, Static analysis remediation, Security hardening
**ML & imaging:** Computer vision, Image registration, Signal processing (FFT/DFT), Hyperspectral imaging, OCR / handwriting recognition, 3D meshes & point clouds, VTK